

Ginkgo

Ginkgo biloba

Species background: Native to a small region of southeastern China; wild ginkgo populations are so rare that the species survived mainly because it was cultivated in temple gardens for over 1,000 years. Ginkgo has no close living relatives at all -- it's the last surviving member of an entire ancient plant lineage that otherwise died out over 200 million years ago.

This packet contains one full lesson for each grade band: K-2, 3-5, 6-8, and 9-12. Each includes a learning objective, standards connection, materials, teacher background, a step-by-step procedure, discussion questions, an assessment, and an extension activity.

GRADE BAND K-2

Ginkgo: Identify ginkgo by its unique fan-shaped leaf, unlike any other tree leaf.

Standards connection: NGSS K-LS1-1; National Core Arts Standards VA:Cr1 (K-2)

Materials: Ginkgo coloring sheet, a real or paper fan, gold/yellow crayons.

Teacher background

Ginkgo leaves are shaped like a fan and no other common tree has this exact shape. In fall, the leaves can all turn gold and drop within a day or two.

Procedure

- 1 Engage: Show a real or paper fan. Ask: what tree leaf looks like this?
- 2 Explore: Compare the ginkgo coloring sheet to 2-3 other leaf shapes (oak, maple). Sort which is the "odd one out."
- 3 Explain: Explain that ginkgo is the only common tree with this fan shape, and that it's an ancient kind of tree.
- 4 Art: Fold a paper fan shape, then color the ginkgo coloring sheet gold.
- 5 Evaluate: Student can pick the ginkgo leaf out of a set of 3 different leaf pictures.

Discussion questions

What other things in nature look like fans?

Assessment

Correctly points to the ginkgo leaf shape among distractors.

Extension

Draw a nighttime scene where all the "stars" are tiny ginkgo fans.

Ginkgo: Explain what a "living fossil" is and why ginkgo earns that name.

Standards connection: NGSS 3-LS4; National Core Arts Standards VA:Cr2 (3-5)

Materials: Coloring sheet, a paper timeline strip, pencil.

Teacher background

Ginkgo trees alive today are nearly identical to fossil ginkgo leaves over 200 million years old -- this tree shape has barely changed since before most dinosaurs existed.

Procedure

- 1 Engage: Ask: if you found this exact leaf shape as a fossil, how old do you think it might be?
- 2 Explore: On a blank timeline strip, mark "Today" and "200 million years ago," then place a ginkgo leaf drawing near the far end.
- 3 Explain: Introduce "living fossil" and explain ginkgo has no close living relatives.
- 4 Art: Fold and color the fan shape gold; sketch one other organism from the same era for scale.
- 5 Evaluate: Student writes one sentence defining "living fossil" in their own words.

Discussion questions

Why do you think ginkgo survived so long when so many relatives went extinct?

Assessment

Correct written definition of "living fossil" using ginkgo as the example.

Extension

Research one other living fossil (horseshoe crab, coelacanth) and compare.

Ginkgo: Explain dioecy (separate male/female trees) in ginkgo and why cities intentionally plant only male trees.

Standards connection: NGSS MS-LS1 (structure and reproduction); National Core Arts Standards VA:Cr2/Re7 (6-8)

Materials: Coloring sheet, research notebook or short reading.

Teacher background

Ginkgo trees are dioecious -- individual trees are either male or female, unlike most trees. Female trees produce a fleshy, foul-smelling seed coating, so nearly all city-planted ginkgos are cloned male trees.

Procedure

- 1 Engage: Why might a city deliberately plant only "one sex" of a tree species -- what tradeoff does that suggest?
- 2 Explore: Research plant dioecy vs. the more common "perfect flower" arrangement (both sexes in one flower).
- 3 Explain: Discuss how cities propagate all-male ginkgo cultivars through grafting, not seed.
- 4 Art: Create a botanical print of a ginkgo leaf using ink and a real fallen leaf, capturing the parallel vein pattern.
- 5 Evaluate: Write a short paragraph weighing the tradeoff -- convenience vs. reduced genetic diversity in city ginkgo populations.

Discussion questions

What might happen to city ginkgo populations long-term if they're all clones of a few male trees?

Assessment

Paragraph correctly explains dioecy and identifies at least one tradeoff of all-male planting.

Extension

Identify another dioecious species and compare its reproductive strategy.

Ginkgo: Evaluate evolutionary conservatism in ginkgo and the genetic risks of clonal urban planting.

Standards connection: NGSS HS-LS4 (evolution); National Core Arts Standards Proficient-level Creating/Responding (9-12)

Materials: Coloring sheet, access to basic research sources.

Teacher background

Ginkgo's genome and morphology have changed remarkably little over roughly 200 million years -- an example of evolutionary stasis. Modern urban ginkgo populations are propagated almost entirely via grafted male clones, creating very low genetic diversity compared to wild or historic populations.

Procedure

- 1 Engage: Does a species that "stops evolving" for 200 million years suggest it's especially fit, especially lucky, or both?
- 2 Explore: Research evolutionary stasis and identify at least one hypothesis for ginkgo's low morphological change.
- 3 Explain: Connect to modern clonal propagation and the general genetic-diversity risk of monoculture plantings (echoing Dutch elm disease's spread through elms).
- 4 Art: Produce a labeled botanical illustration plate with scale bar and observational caption, capturing the parallel venation pattern.
- 5 Evaluate: Write a short argument -- should cities diversify ginkgo cultivars, or is the tradeoff worth it? Support with evidence.

Discussion questions

Is "evolutionary success" the same as "genetic diversity"? Can a species be evolutionarily successful and still vulnerable today because of how humans propagate it?

Assessment

Argument essay evaluated on use of evidence and clarity of reasoning.

Extension

Compare ginkgo's clonal-monoculture risk to the American elm / Dutch elm disease case study.

SOURCES & METHODOLOGY

Background research drawn from TreeWalk NYC's own species research. Lesson structure (objective, background, materials, procedure, assessment, extension) follows the format used by real, freely published environmental-education lessons, including the National Park Service's "Looking at Leaves" 5E lesson plan (Acadia Outdoor Classroom Collaborative), Penn State Extension's "Sorting and Classifying Leaves," and Project Learning Tree's leaf-activity guidance -- adapted and written fresh for this species, not copied from those sources. This is an educational pilot; standards references indicate topic alignment, not formal certification.